Inference Server
Network Application
NEFSim
MQTT Broker

par [QoS monitoring and adaptation]

loop [every repPeriod]

1. POST {notifUri}
AnalyticsEventNotification [notifId, analyEventNotifs (wlanInfos)]

2. 204 No Content

3. Select the camera entry (gpsi) and compare uplinkRate and downlinkRate with the minimum UL/DL bitrates

opt [KPI threshold breached and no downgrade applied]

loop [for each pedestrian and vehicle UE]

4. GET /3gpp-as-session-with-qos/v1/{scsAsId}/subscriptions

5. 200 OK [AsSessionWithQoSSubscription list]

alt [UE subscription exists]

6.a. PATCH /3gpp-as-session-with-qos/v1/{scsAsId}/subscriptions/{subscriptionId}
[qosReference = downgrade profile]

7.a. 200 OK

6.b. POST /3gpp-as-session-with-qos/v1/{scsAsId}/subscriptions
[notificationDestination, ueIpv4Addr | gpsi, dnn, qosReference = downgrade profile]

7.b. 201 Created [AsSessionWithQoSSubscription (self)]

8. POST {notificationDestination}
UserPlaneNotificationData [transaction, eventReports
(event = SUCCESSFUL_RESOURCES_ALLOCATION | FAILED_RESOURCES_ALLOCATION)]

9. 204 No Content

[Pedestrian tracking]

loop [on every pedestrian location update]

10. POST {notificationDestination}
MonitoringNotification [monitoringEventReports (locationInfo.geographicArea.point)]

11. 204 No Content

12. Evaluate the point against the surveyed area; on a state transition, add or remove the pedestrian from the camera area set

[Collision detection]

loop [for each processed frame]

13. POST /callbacks/crash_inference/{cameraSupi}
[detections_number, predictions, frame_number]

14. 204 No Content

15. Filter by class and confidence threshold; confirm persistence across the frame window

opt [collision confirmed]

16. POST /3gpp-monitoring-event/v1/{scsAsId}/subscriptions
[monitoringType = LOCATION_REPORTING, maximumNumberOfReports = 1, msisdn | ipv4Addr]

17. 201 Created [locationInfo (geographicArea (camera location))]

par

loop [for each pedestrian UE in the camera area set]

18. POST /api/v1/sms/send [gpsi, text]

19. 200 OK

20. PUBLISH {topicBase}/{stationId}/DENM
[header, denm (management (eventPosition, relevanceDistance, validityDuration), situation (eventType = accident))]

21. Relay the DENM to the vehicles subscribed to the event area